A HOMOTOPY COMMUTATIVE SQUARE INDUCES A MORPHISM OF THE ASSOCIATED TRIANGLES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-cone (source lines 3778-3798, which also contains the author's slogan environment naming the triangulated-category axioms it serves); the proof environment begins two lines below it and is its own proof, closing at source line 3830.

Editorial note. The statement refers to the author's standing Situation situation-ABC by label; that situation's text is retained as a context span placed before the counted statement, and the generator expands the context transitively, so the reference resolves inside the assembled package.

Editorial note. The statement contains one Xy-pic square and one displayed triangle; both are part of the retained source range and are reproduced as drawn, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: a square commuting up to homotopy induces a morphism of the associated triangles, an axiom of the triangulated structure.

Allowed standard prerequisites (editorial).

- The standing situation (axioms A, B, C). A dg category with the distinguished-triangle axioms, in particular the admissible short exact sequence attached to a morphism and its associated triangle (the author's Situation situation-ABC, retained as context).. It supplies the two admissible sequences, their structure maps and the associated triangles (source lines 3784-3798, 3800-3810).
- Cone of a morphism and its structure maps. The cone c(f) of a morphism with the maps realising it as the middle term of the admissible sequence, and the identities among those maps and their differentials.. The component c is built out of those maps and verified using the identities (source lines 3805-3827).
- Commuting up to homotopy. The given square commutes up to homotopy, that is, the two composites differ by the differential of a homotopy.. The homotopy is one of the ingredients of the matrix defining c and is what makes the computation work (source lines 3786-3791, 3811-3826).

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Situation 0.1. Here R is a ring and $\mathcal{A}$ is a differential graded category over R having axioms (A), (B), and

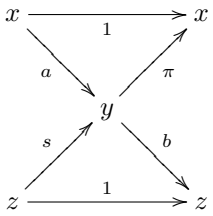
(C) given an arrow $f : x \to y$ of degree 0 with $\mathrm{d}(f) = 0$ there exists an admissible short exact sequence $y \to c(f) \to x[1]$ in $\mathrm{Comp}(\mathcal{A})$ such that the map $x[1] \to y[1]$ of Lemma 0.2 is equal to $f[1]$.

Lemma 0.2. *Let $\mathcal{A}$ be a differential graded category satisfying axioms (A) and (B). Given an admissible short exact sequence $x \to y \to z$ we obtain (see proof) a triangle*

$$x \to y \to z \to x[1]$$

in Comp($\mathcal{A}$) with the property that any two compositions in $z[-1] \to x \to y \to z \to x[1]$ are zero in $K(\mathcal{A})$.

Proof. Choose a diagram



giving the isomorphism of graded objects $y \cong x \oplus z$ as in the definition of an admissible short exact sequence. Here are some equations that hold in this situation

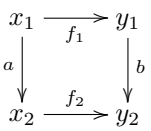
(1) $1 = \pi a$ and hence $\mathrm{d}(\pi)a = 0$,
(2) $1 = bs$ and hence $b\mathrm{d}(s) = 0$,
(3) $1 = a\pi + sb$ and hence $a\mathrm{d}(\pi) + \mathrm{d}(s)b = 0$,
(4) $\pi s = 0$ and hence $\mathrm{d}(\pi)s + \pi\mathrm{d}(s) = 0$,
(5) $\mathrm{d}(s) = a\pi\mathrm{d}(s)$ because $\mathrm{d}(s) = (a\pi + sb)\mathrm{d}(s)$ and $b\mathrm{d}(s) = 0$,
(6) $\mathrm{d}(\pi) = \mathrm{d}(\pi)sb$ because $\mathrm{d}(\pi) = \mathrm{d}(\pi)(a\pi + sb)$ and $\mathrm{d}(\pi)a = 0$,
(7) $\mathrm{d}(\pi\mathrm{d}(s)) = 0$ because if we postcompose it with the monomorphism a we get $\mathrm{d}(a\pi\mathrm{d}(s)) = \mathrm{d}(\mathrm{d}(s)) = 0$, and
(8) $\mathrm{d}(\mathrm{d}(\pi)s) = 0$ as by (4) it is the negative of $\mathrm{d}(\pi\mathrm{d}(s))$ which is 0 by (7).

We've used repeatedly that $\mathrm{d}(a) = 0$, $\mathrm{d}(b) = 0$, and that $\mathrm{d}(1) = 0$. By (7) we see that

$$\delta = \pi\mathrm{d}(s) = -\mathrm{d}(\pi)s : z \to x[1]$$

is a morphism in $\mathrm{Comp}(\mathcal{A})$. By (5) we see that the composition $a\delta = a\pi\mathrm{d}(s) = \mathrm{d}(s)$ is homotopic to zero. By (6) we see that the composition $\delta b = -\mathrm{d}(\pi)sb = \mathrm{d}(-\pi)$ is homotopic to zero. □

Lemma 0.3. *In Situation 0.1 suppose that*



is a diagram of Comp($\mathcal{A}$) commutative up to homotopy. Then there exists a morphism $c : c(f_1) \to c(f_2)$ which gives rise to a morphism of triangles

$$(a, b, c) : (x_1, y_1, c(f_1)) \to (x_1, y_1, c(f_1))$$

in $K(\mathcal{A})$.

Proof. The assumption means there exists a morphism $h : x_1 \to y_2$ of degree -1 such that $\mathrm{d}(h) = bf_1 - f_2a$. Choose isomorphisms $c(f_i) = y_i \oplus x_i[1]$ of graded objects compatible with the morphisms $y_i \to c(f_i) \to x_i[1]$. Let's denote $a_i : y_i \to c(f_i)$, $b_i : c(f_i) \to x_i[1]$, $s_i : x_i[1] \to c(f_i)$, and $\pi_i : c(f_i) \to y_i$ the given morphisms. Recall that $x_i[1] \to y_i[1]$ is given by $\pi_i \mathrm{d}(s_i)$. By axiom (C) this means that

$$f_i = \pi_i \mathrm{d}(s_i) = -\mathrm{d}(\pi_i)s_i$$

(we identify $\mathrm{Hom}(x_i, y_i)$ with $\mathrm{Hom}(x_i[1], y_i[1])$ using the shift functor [1]). Set $c = a_2b\pi_1 + s_2ab_1 + a_2hb$. Then, using the equalities found in the proof of Lemma 0.2 we obtain

$$\begin{aligned} \mathrm{d}(c) &= a_2b\mathrm{d}(\pi_1) + \mathrm{d}(s_2)ab_1 + a_2\mathrm{d}(h)b_1 \\ &= -a_2bf_1b_1 + a_2f_2ab_1 + a_2(bf_1 - f_2a)b_1 \\ &= 0 \end{aligned}$$

(where we have used in particular that $\mathrm{d}(\pi_1) = \mathrm{d}(\pi_1)s_1b_1 = f_1b_1$ and $\mathrm{d}(s_2) = a_2\pi_2\mathrm{d}(s_2) = a_2f_2$). Thus c is a degree 0 morphism $c : c(f_1) \to c(f_2)$ of $\mathcal{A}$ compatible with the given morphisms $y_i \to c(f_i) \to x_i[1]$. □